

GIMLI AP, MB, Canada

WMO: 717480

Lat: 50.6394N

Lon: 97.0460W

Elev: 230

StdP: 98.59

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-30.4	-37.6	0.1	-33.0	-34.7	0.1	-30.2	12.9	-12.1	11.6	-13.4	2.4	280	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	28.9	20.8	27.2	19.7	25.8	18.8	22.4	26.9	21.2	25.5	20.1	24.2	5.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	16.0	25.1	19.6	14.7	23.9	18.5	13.7	22.6	67.3	27.0	62.6	25.6	58.7	24.3	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	10.0	8.8	DB	-37.4	32.5	2.9	2.1	-39.5	34.0	-41.2	35.2	-42.8	36.4	-44.9	38.0	(4)
(5)			WB	-37.5	24.7	3.0	1.6	-39.6	25.8	-41.4	26.7	-43.0	27.6	-45.2	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-16.7	-15.0	-6.9	2.6	9.8	16.3	18.9	17.8	12.8	5.1	-4.5	-13.4	(6)
(7)		DBStd	13.99	8.26	7.70	7.86	6.07	4.88	3.87	3.07	3.34	4.40	4.71	6.64	7.89	(7)
(8)		HDD10.0	3692	829	699	523	228	64	3	0	0	22	164	434	725	(8)
(9)		HDD18.3	5953	1087	933	781	471	266	82	31	52	173	410	684	983	(9)
(10)		CDD10.0	893	0	0	1	7	59	192	276	241	105	13	0	0	(10)
(11)		CDD18.3	112	0	0	0	0	2	21	48	34	7	0	0	0	(11)
(12)		CDH23.3	1027	0	0	0	2	49	197	407	309	61	2	0	0	(12)
(13)		CDH26.7	207	0	0	0	0	9	42	79	69	8	0	0	0	(13)
(14)	Wind	WSAvg	4.1	4.2	4.1	4.2	4.3	4.5	4.0	3.6	4.0	4.4	4.4	4.0	(14)	
(15)	Precipitation	PrecAvg	529	21	16	28	35	62	86	78	68	52	37	25	25	(15)
(16)		PrecMax	766	50	36	77	56	109	184	148	116	115	79	75	65	(16)
(17)		PrecMin	339	5	6	3	17	25	37	40	20	17	5	7	6	(17)
(18)		PrecStd	113	14	10	21	12	29	44	27	28	27	22	15	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	3.7	13.2	22.3	27.8	30.6	31.0	31.1	28.0	21.3	12.5	3.4	(19)
(20)		MCWB	1.5	1.7	7.7	12.6	15.9	21.9	23.1	21.9	19.0	14.5	8.1	1.2	(20)	
(21)		2%	DB	0.4	1.5	8.1	18.2	24.4	27.4	28.7	28.6	25.1	17.8	8.6	0.7	(21)
(22)		MCWB	-0.7	0.5	4.8	9.8	14.3	19.1	21.5	20.6	17.7	12.2	5.7	-0.4	(22)	
(23)		5%	DB	-2.3	-0.1	5.0	14.8	21.5	25.5	27.2	26.6	22.7	15.0	6.1	-0.8	(23)
(24)		MCWB	-3.1	-0.9	2.9	8.4	13.2	17.9	20.4	19.8	16.4	10.8	4.0	-1.6	(24)	
(25)		10%	DB	-4.7	-3.2	2.9	11.9	18.6	23.7	25.7	24.8	20.4	12.5	3.7	-2.5	(25)
(26)		MCWB	-5.4	-4.0	1.3	6.8	12.0	17.3	19.5	18.8	15.6	9.3	2.2	-3.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	2.0	9.2	13.3	18.2	23.6	24.7	23.3	20.9	15.4	8.8	1.4	(27)
(28)		MCDWB	2.4	3.0	12.1	20.2	24.4	29.3	29.5	28.1	25.3	20.3	11.4	2.9	(28)	
(29)		2%	WB	-0.8	0.7	5.2	10.7	15.9	20.9	22.9	21.8	18.9	13.2	6.2	-0.2	(29)
(30)		MCDWB	0.2	1.5	7.8	17.4	21.3	25.3	27.1	26.7	22.5	16.6	8.4	0.6	(30)	
(31)		5%	WB	-3.2	-1.1	3.0	8.9	14.2	19.3	21.5	20.7	17.6	11.3	4.1	-1.7	(31)
(32)		MCDWB	-2.4	0.0	5.0	13.8	19.2	23.6	25.7	25.3	21.2	14.4	6.1	-0.9	(32)	
(33)		10%	WB	-5.4	-3.9	1.4	7.3	12.9	18.1	20.4	19.7	16.2	9.4	2.3	-3.3	(33)
(34)		MCDWB	-4.7	-3.1	2.8	11.8	18.0	22.1	24.4	23.7	19.8	12.4	3.8	-2.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.7	10.6	10.3	10.9	11.5	10.5	11.0	11.4	10.8	8.7	7.8	8.7	(35)	
(36)		5% DB	MCDBR	9.9	9.1	10.2	15.7	16.5	13.5	12.9	13.8	14.3	13.0	10.6	9.1	(36)
(37)		MCWBR	9.2	8.2	7.2	9.3	8.5	6.9	7.1	7.3	7.9	8.1	7.7	8.2	(37)	
(38)		5% WB	MCDBR	10.0	9.2	9.5	14.5	12.9	11.2	11.6	12.5	11.7	12.0	9.8	9.0	(38)
(39)		MCWBR	9.3	8.3	7.0	9.0	7.7	6.6	7.1	7.2	7.3	7.9	7.3	8.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.263	0.296	0.306	0.354	0.368	0.384	0.393	0.377	0.346	0.300	0.271	0.247	(40)	
(41)		taud	2.223	2.136	2.278	2.334	2.371	2.355	2.357	2.405	2.479	2.559	2.423	2.289	(41)	
(42)		Ebn at Noon	765	823	897	888	888	873	857	856	849	838	770	739	(42)	
(43)		Edh at Noon	70	101	106	114	115	118	116	105	88	67	58	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.22	2.39	3.67	5.04	5.45	5.79	6.03	5.01	3.55	2.06	1.10	0.88	(44)	
(45)		RadStd	0.12	0.17	0.27	0.34	0.46	0.46	0.28	0.35	0.32	0.26	0.16	0.09	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-0.70	-0.96	N/A	N/A	N/A	N/A	(46)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	-0.71	-0.99	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page